SCIENCE

How to become a lightning calculator

Anonymous

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Transcriber’s Notes:

Text enclosed by underscores is in italics (_italics_).

The whole number part of a mixed fraction is separated from the fractional part with -, for example, 2-1/2.

Additional Transcriber’s Notes are at the end.

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How to Become a LIGHTNING CALCULATOR.

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Instantaneous Addition.

Accuracy should be first considered, then rapidity. Quick adders, by the way, are the most accurate. Write the numbers in vertical lines, avoiding irregularity. This is important. Keep your thought on results not numbers themselves. Do not reckon 7 and 4 are 11 and 8 are 19, but say 7, 11, 19 and so on.

When the same number is repeated several times, multiply instead of adding.

When adding horizontally begin at the left.

3132 2453 12 | 6471 20 | 7312 15 | 2134 21 | ----- ---+ 21502

In adding long columns, prove the work, by adding each column separately in the opposite direction, before adding the next column. Many accountants put down both figures as in the illustration. The sum of the first column is 12; carrying _one_, the sum of the second is 20; carrying _two_, the sum of the third column is 15; carrying _one_, the sum of the fourth column is 21, and the total, 21502, is found by calling off the last two figures and the right-hand figures, following the wave line in the illustration. This method is better than the old one of penciling down the number _to carry_. If one desires to go back and add a certain column a second time, the number to carry is at hand and the former total is known.

How to Add Two Columns at Once.

2312 3253 2610 1256 3199 ----- 12630

To the inexperienced it will be a difficult task to add two columns at once, but many of those who have daily practice in addition find it about as easy to add two columns as one. Say 99 and 50 are 149, and 6 are 155, and 10 and 50 are 215 and 3 are 218, and 12 are 230. Carry 2, and say 33 and 12 are 45, and 20 are 65, and 6 are 71, and 30 are 101, and 2 are 103, and 23 are 126.

Much of the information here contained is compiled from W. D. Rowland’s valuable little volume, entitled “How to become expert with figures.” You can get this handy book by sending 25 cents in stamps to AMERICAN NATION CO., Boston.

Multiplication.

To Multiply Any Number by 11.

Write the first right-hand figure, add the first and second, the second and third, and so on; then write the left-hand figure. Carry when necessary.

219434 × 11 = 2413774

Put down the right-hand figure 4. Then say, 4 and 3 are 7; then, 3 and 4 are 7; then, 4 and 9 are 13, put down 3 and carry 1; then, 9 and 1 and 1 are 11, put down the 1 and carry 1; then, 1 and 2 and 1 are 4; then write the left-hand figure 2. In multiplying small numbers, such as 24 by 11, write the sum of the two figures between the two figures, making 264, the required product.

To Multiply by 101, 1001, etc.

To multiply by 101, add two ciphers to the multiplicand, and add to this the multiplicand.

2341 × 101 = 234100 + 2341

To multiply by 1001, add three ciphers to the multiplicand, and add to this the multiplicand.

To Multiply by 5, 25, 125.

To multiply by 5, add a cipher and divide by 2.

To multiply by 25, add two ciphers and divide by 4.

To multiply by 125, add three ciphers and divide by 8.

Another Easy Way to Multiply.

82 54 ---- 4428

To multiply two figures by two figures, proceed as follows: Multiply units by units for the _first_ figure.

Carry and multiply tens by units and units by tens, (adding) for the _second_ figure. Carry and multiply tens by tens for the _remaining_ figure or figures. In this example proceed as follows:

2 × 4 = 8 = 1st figure. (4 × 8) + (5 × 2) = 42. Therefore 2 = 2d figure. (5 × 8) + 4 carried = 44 = 3d and 4th figures.

By a little practice any one may become as familiar with this rule and as ready in its application as with the ordinary method.

To multiply any number by 2-1/2, add one cipher, and divide by 4.

To multiply any number by 3-1/3, add one cipher, and divide by 3.

To multiply by 33-1/3, add two ciphers, and divide by 3.

To multiply any number by 1-3/7, add one cipher, and divide by 7.

To multiply by 16-2/3, add two ciphers, and divide by 6.

To multiply by 14-2/7, add two ciphers, and divide by 7.

To multiply by 875, add three ciphers, and divide by 8.

To divide by 25, multiply by 4, and cut off two figures.

To divide by 125, multiply by 8, and cut off three figures.

To multiply by 12-1/2, add two ciphers, and divide by 8.

To find the value of any number of articles at 75 cents each, deduct one-quarter of the number from itself and call the remainder dollars.

To Subtract Any Number Consisting of Two Figures from 100.

Take the first figure from 9, and the second from 10. For example: in subtracting 73 from 100, or in taking 73 cents change out of a dollar, say 7 from 9 and 2, and 3 from 10 and 7, or 27 cents. Practice this rule. It is simple, and will be found particularly helpful in making change.

Divisions.

A number is divisible by 2 when the last digit is even.

A number is divisible by 4 when the last two digits are divisible by 4.

To divide by 12-1/2, multiply by 8, and cut off two figures.

Simple Discount Rule.

This simple rule is in use in many houses where several discounts are allowed from list prices. Suppose the list price of a piano to be $500, and you allow an agent 25, 20 and 10 off.

100 100 100 25 20 10 --- --- --- 75 × 80 × 90 = .540000

Subtract each from 100 and multiply and you get .54. $500 × .54 = $270, the agent’s price.

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If you want to get a complete book on quick calculating, comprising all modern methods, together with a vast amount of other valuable matter, pertaining to business, send 25 cents to AMERICAN NATION CO., Boston, Mass., for a volume of “How to Become Expert at Figures.”

Percentage.

The name percentage is applied to certain arithmetical exercises in which 100 is used as the basis of computation. Per cent. is an abbreviation of the Latin _per centum_, meaning _by the hundred_. This sign % is used for the words _per cent._ Thus, 10% of a number equals 10/100, or 1/10 of the number; 50% equals 50/100, or

1/2, etc.

FRACTIONAL EQUIVALENTS.

50 % = .50 = 1/2. 33-1/3 % = .33-1/3 = 1/3. 25 % = .25 = 1/4. 20 % = .20 = 1/5. 16-2/3 % = .16-2/3 = 1/6. 12-1/2 % = .12-1/2 = 1/8. 10 % = .10 = 1/10. 8-1/3 % = .08-1/3 = 1/12. 6-1/4 % = .06-1/4 = 1/16. 5 % = .05 = 1/20. 2-1/2 % = .02-1/2 = 1/40.

Accurate Interest.

Interest is the sum charged for the use of money. It is really the _use of money_ or the benefit derived from its use. The principal is the sum for the use of which interest is paid. The rate of interest is the _per cent._ of the principal charged for its use for _one year_. Simple interest is the interest on the principal only, for the full time; compound interest is interest on the principal for the full time, and interest on each interest payment after it becomes due.

To find the accurate interest on any sum of money at a given rate for one year, multiply the sum by the rate and divide by 100.

To find the accurate interest on any sum of money at a given rate for any given number of days, multiply the interest for one year by the number of days and divide the product by 365.

Equation of Payments.

Equation of Payments is the process of finding when two or more sums due at different times may be paid at once, without loss to debtor or creditor. The time for such payment is called the equated time.

To equate two or more payments, multiply each payment by its time, and divide the sum of the products by the sum of the payments.

The times of the several payments must be in the same denomination, and this will be the denomination of the answer.

Less than 1/2 day is rejected; 1/2 day or more counts as 1 day. If the date is required, reckon the equated time forward from the given date.

The Lightning Calculator’s Addition.

4379321 5620679 2184509 7815491 2105610 3453173 -------- 25558783

There are experts who can add very rapidly. The best of them, however, cannot add up a column of _ones_ any faster than you can. Here is how some of the rapid addition is performed. The operator writes a line of figures, then another, and so on. The second line, however, added to the first makes _nines_, except at the extreme right, where the two figures add to _ten_. The third and fourth bear the same relation, and as many more as he chooses to put down. The last two lines, however, are put down at random. Now, to add these columns, he begins anywhere, perhaps at the left-hand side, putting down 2 (the number of _pairs_ above), then by simply adding the two bottom lines, he gets the correct sum.

Bank Discount.

The sum charged by a bank for cashing a note or time draft is called bank discount. This _discount_ is the simple interest, paid in advance, for the number of days the note has to run. Wholesale business houses usually sell goods on time and take notes from the retailers in payment. These notes are not often for a longer period than three months. Some are placed in the banks for _collection_, others are discounted. When a note is discounted at a bank the _payee_ endorses it, making it payable to the bank. Both maker and payee are then responsible to the bank for its payment. If the note is drawing interest the discount is reckoned on and deducted from the amount due at maturity. Most notes discounted at banks do not draw interest. The time in bank discount is always the number of days from the _date of discounting_ to the _date of maturity_.

Example. A note of $250, dated July 7th, payable in 60 days, is discounted July 7th, at 6 per cent.; find the proceeds.

This note is due in 63 days, or September 8th. The accurate interest of $250 for 63 days at 6 per cent. is $2.59. The proceeds, then, will be $250 - $2.59, or $247.41.

How to Make Change.

Salesmen usually make change by addition. They have the money to count out, and in doing so they add to the amount of the purchase until they reach the amount of the bill presented. For example, if you buy something worth $3.35 and present a ten-dollar bill in payment, you will probably receive in return 5 cents, 10 cents, 50 cents, $1, and $5; the salesman saying 40, 50, $4, $5; $10. This method is least liable to error.

Accuracy and rapidity in counting out change can best be acquired by practice behind the counter or at the cash-desk.

Proof of Multiplication in Ten Seconds.

The _unitate_ of a number is the sum of its digits reduced to a unit.

252 = 9 } = 54 = 9 321 = 6 } --- 252 504 756 ----- 80892 = 27 = 9

The _unitate_ of the multiplier is 9 and the _unitate_ of the multiplicand is 6; 6 times 9 equals 54, and the _unitate_ of 54 is 9. Now the _unitate_ of the product is found to be 9 also, which is a proof of the correctness of the work.

The Canadian Interest Rule.

4 | 724 181 × 11 = 19.91

This rule of computing interest appears in some Canadian text-books, and, though simply a modification of other rules, is worthy of notice. To find the interest on $724 for 5-1/2 months at 6 per cent., all you have to do is to divide by 4 and multiply by 11. The rule is to divide the principal by 4, and to multiply the quotient by one-third of the product of the rate by the time in months. _Six_ times 5-1/2 = 33, and one-third of 33 is 11. If the time be expressed in years, multiply one-fifth of the principal by one-half the product of the rate by the number of years, and remove the decimal point one place to the left.

Table of Transposed Numbers.

DIFFERENCES.

9 { 10 21 32 43 54 65 76 87 98 { 01 12 23 34 45 56 67 78 89

18 { 20 31 42 53 64 75 86 97 { 02 13 24 35 46 57 68 79

27 { 30 41 52 63 74 85 96 { 03 14 25 36 47 58 69

36 { 40 51 62 73 84 95 { 04 15 26 37 48 59

45 { 50 61 72 83 94 { 05 16 27 38 49

54 { 60 71 82 93 { 06 17 28 39

63 { 70 81 92 { 07 18 29

72 { 80 91 { 08 19

81 { 90 { 09

90 { 100 { 010

99 { 100 { 001

Explanation of Foregoing Table.

The transposition of figures is a frequent cause of errors in proving accounts and balance sheets. This table is founded on the fact that all differences between transposed numbers are multiples of nine. The difference between the figures misplaced is equal to the quotient of the resulting error when divided by nine; thus, 91 - 19 = 72; 72 ÷ 9 = 8; 9 - 1 = 8, and the labor of searching for it may be confined to examining those figures the transposition of which would make the difference, as they are the only ones that can cause the error. Thus: if the error in the balance-sheet be 81 cents, it is possibly caused by a transposition, and the clerk can first examine the cents column of his books for items of 90 cents, or 09 cents, alone, with a strong probability of finding the cause of the error without further revision. Transpositions may occur in any decimal or integer place, and the differences caused thereby are divisible by nine without a remainder; but, beyond this table, the numbers ascend in regular progression, each difference increasing by nine, as follows:

108 { 120; 117 { 130; 126 { 140; etc. { 12 { 13 { 14

The quotient of the difference in a regular progression, when divided by nine, gives the figures transposed, thus: 130 - 13 = 117 ÷ 9 = 13, which are the figures to be sought for when a discrepancy of 117 is shown; but this will not apply to differences below 81, nor to mixed transpositions. An error divisible by two may be caused by posting an item to the wrong side of the ledger.

Short Method to Find the Interest of a Given Sum.

Reduce the time to months, and to the number thus found annex one-third of the days, which whole number multiplied by one-half of your principal will produce you the required interest in dollars, cents and mills, at 6 per cent. If days only are given, multiply one-third of the days by one-half of the principal for the required interest at

6 per cent. Note these exercises:

$250 at 6% for 8 mos. 6 ds. = 82 × 125 = $10.25.

$250 at 6% for 93 ds. = 31 × 125 = $3.88.

$250 at 6% for 9 mos. = 9 × 125 = $11.25.

In some respects this rule is superior to the well-known 60-day method of reckoning interest.

Interest Computations.

462.50 .48 ------- 6 | 360 ---- 60 | 222.0000 | 3.70

Multiply the _principal_ (amount of money at interest) by the time, reduced to days; then divide this product by the quotient obtained by dividing 360 (the number of days in the interest year) by the per cent. of interest, and the quotient thus obtained will be the required interest. Require the interest of $462.50 for one month and eighteen days at 6 per cent. An interest month is 30 days; one month and 18 days equals 48 days. $462.50 multiplied by .48 gives $222.0000; 360 divided by 6 (the per cent. of interest) gives 60, and $222.0000 divided by 60 will give you the exact interest, which is $3.70. If the rate of interest in the above example were 12 per cent., we would divide the $222.0000 by 30 (because 360 divided by 12 gives 30); if 4 per cent., we would divide by 90; if 8 per cent., by 45; and in like manner for any other per cent.

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Transcriber’s Notes:

Use of - to represent division in some expressions is standardized to /.

The following change was made:

p. 15: Extraneous numbers were removed from the example interest computation.